THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** solution	Uniform composition.
2	**(c)** brass	Alloy.
3	**(c)** (mass of solute / mass of solution) x 100	Standard definition.
4	**(c)** solute	Substance dissolved.
5	**(a)** nitrogen	Most abundant gas.
6	**(a)** 20 per cent	20/100 x 100.
7	**(c)** solvent	Dissolving medium.
8	**(a)** suspension	Large particles.
9	**(b)** 100 g	Solute + solvent.
10	**(a)** dissolves a large number of substances	It dissolves many but not all.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Solute is dissolved, solvent dissolves it (1); Solution is their homogeneous mixture (1).	1+1
12	Solution is transparent, particles invisible; suspension is cloudy (1); Suspension particles settle and can be filtered (1).	1+1
13	Mass of solution = 50 g (1); (5 / 50) x 100 = 10 per cent (1).	1+1
14	Maximum amount of solute that dissolves in 100 g of solvent at a given temperature (1); Depends on temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add salt/sugar in portions to 100 mL of cold and of warm water (1); Count spoons dissolved till saturation (1); More dissolves in warm water (1).	1+1+1
16	Mass of solute = 10/100 x 200 (1); = 20 g (1); Mass of water = 180 g (1).	1+1+1
17	Drinks and tea (1); Medicines like ORS (1); Cleaning with soap solutions (1).	1+1+1
	OR: Add salt to water a little at a time and stir (1); Continue until no more dissolves (1); Undissolved salt settles: saturated (1).	1+1+1
18	Gas solubility decreases with temperature (1); Carbon dioxide escapes (1); Fizz is lost (1).	1+1+1
	OR: Oil and water are unlike (1); Oil and kerosene are alike (1); Like dissolves like (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Particle size (1); Appearance (1); Settling (1); Filtration (1); Examples (1).	1+1+1+1+1
	OR: Solid in liquid: sugar in water (1); Gas in liquid: soda water (1); Liquid in liquid: alcohol in water (1); Gas in gas: air (1); Solid in solid: brass (1).	1+1+1+1+1
20	Solids: more at high temperature (1); Sugar in hot tea (1); Gases: less at high temperature (1); Aquatic life in warm water (1); Fizz of soda (1).	1+1+1+1+1
	OR: Beam of light passes through a mixture (1); Colloid scatters light, beam visible (1); True solution does not (1); Examples: milk, sugar solution (1); Suspension also scatters but settles (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** suspension	Soil settles.
2	**(a)** sum of the masses of solute and solvent	Conservation of mass.
3	**(a)** saturated	Saturated.
4	**(d)** very small	Tyndall effect not seen.
5	**(b)** copper ions in water	Hydrated Cu2+.
6	**(c)** unsaturated	Unsaturated.
7	**(b)** colloid	Particles scatter light.
8	**(b)** oxygen	Respiration.
9	**(a)** solubility	Definition.
10	**(d)** suspension	Large particles stay on filter.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Solubility generally increases with temperature (1); More solid can dissolve in hot water (1).	1+1
12	Concentrated: more solute (1); Dilute: less solute (1).	1+1
13	Saturated: cannot dissolve more solute (1); Unsaturated: can dissolve more (1).	1+1
14	Oxygen is dissolved in water (1); They use it for respiration (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Homogeneous (1); Particles cannot be seen and do not settle (1); Cannot be separated by filtration (1).	1+1+1
16	Soda water (1); Air (1); Oxygen in water (1).	1+1+1
17	Salt: evaporation (1); Sand: filtration (1); Different types of mixture (1).	1+1+1
	OR: Heterogeneous (1); Particles settle (1); Separated by filtration (1).	1+1+1
18	(a) Salt, water (1); (b) CO2, water (1); (c) Iodine, alcohol (1).	1+1+1
	OR: Salt particles fit in spaces between water particles (1); Little change in volume (1); Shows particles have gaps (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Add salt in portions (1); Stir after each (1); Stop when some stays undissolved (1); Filter and verify (1); Warm it to dissolve more (1).	1+1+1+1+1
	OR: ORS (1); Syrups (1); Salt in cooking (1); Soap solution (1); Dilute vs concentrated (1).	1+1+1+1+1
20	Mass of solution = 200 g (1); Percentage = 12.5 per cent (1); Sugar in 400 g = 0.125 x 400 (1); = 50 g (1); Water = 350 g (1).	1+1+1+1+1
	OR: Add a little more solute (1); Dissolves: unsaturated (1); Does not dissolve: saturated (1); Cooling a saturated solution forms crystals (1); Heating dissolves more (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** decreases with an increase in temperature	Warm water holds less gas.
2	**(a)** increases with temperature	Warmer water dissolves more.
3	**(b)** add more solvent	Dilution.
4	**(d)** dissolved carbon dioxide escapes more	Lower solubility.
5	**(a)** sand	Sand is insoluble.
6	**(c)** a large amount of solute in a given volume	More solute.
7	**(b)** speeds up dissolving	Faster contact.
8	**(c)** soda water	Carbon dioxide in water.
9	**(d)** cannot be seen even with an ordinary microscope	They are extremely small.
10	**(a)** increases the rate of dissolving	More surface area.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Air (1); Mixture of gases in a cylinder (1).	1+1
12	Dissolves many substances (1); Even though not all (1).	1+1
13	Stirring (1); Heating or crushing the solute (1).	1+1
14	Dissolved carbon dioxide escapes (1); Gas solubility falls at low pressure (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Colloid particles scatter light; solution's do not (1); Colloid is more turbid (1); Both do not settle quickly (1).	1+1+1
16	Brings fresh solvent to the solute (1); Increases speed of dissolving (1); Does not increase the amount that dissolves (1).	1+1+1
17	Add more solute (1); Evaporate some solvent (1); Cool the solution (1).	1+1+1
	OR: Mass of solution = 150 g (1); (30/150) x 100 (1); = 20 per cent (1).	1+1+1
18	Larger surface area (1); More contact with water (1); Rate of dissolving increases (1).	1+1+1
	OR: Distilled water does not conduct (1); Tap water has dissolved salts (1); Conducts a little (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Stirring, temperature, particle size (1); Same sugar and water amounts (1); Vary only temperature (1); Measure time taken (1); Hot water is fastest (1).	1+1+1+1+1
	OR: Prepare hot saturated solution (1); Filter (1); Cool slowly (1); Crystals form (1); Dry them (1).	1+1+1+1+1
20	Transports nutrients in blood (1); Dissolves gases for fish (1); Cell reactions (1); Plants absorb minerals (1); Cleaning and cooking (1).	1+1+1+1+1
	OR: More gas dissolves under pressure (1); On opening, pressure falls (1); Gas escapes and fizzes (1); Cool to retain gas (1); Warm drinks lose gas faster (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.